Molecular Dynamics of Neuronal Membranes and Ion Channels Under Ultrasound and Laser Stimulation: Implications for Alzheimer's Disease

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Abstract

Altered membrane mechanics and ion channel dysfunction are key features of Alzheimer's disease (AD) pathology. In this talk, I will present molecular dynamics simulations that reveal how differences in membrane elasticity between healthy and AD neurons affect amyloid-beta (Aβ) interactions and the structural stability of ion channels.

The first part will focus on equilibrium membrane properties, showing how reduced elasticity in AD membranes enhances Aβ binding and destabilizes channel conformation. In the second part, I will discuss how laser and ultrasound stimulation modulate membrane structure and ion transport. Specifically, laser can dissociate ion channels, while ultrasound induces pore formation and regulates ion current, offering a potential route for drug delivery and neuromodulation in AD treatment.

These findings provide molecular-level insights into the altered biophysical properties of AD neuronal membranes and the therapeutic potential of non-invasive physical stimuli.